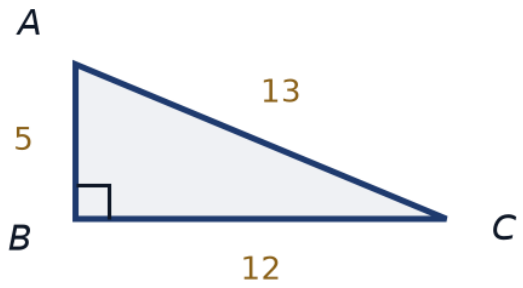
Right-Triangle Trigonometry

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Trigonometric ratios

A right triangle has legs of length 5 and 12 and hypotenuse 13. For the angle θ opposite the side of length 5, find:



a $\sin\theta$ **b** $\cos\theta$ **c** $\tan\theta$

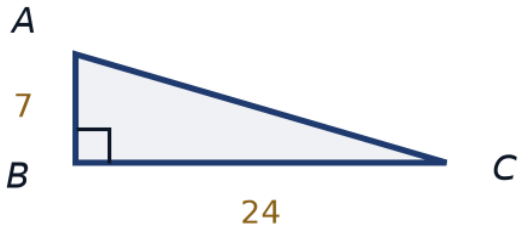
2 [Foundational] • Target: Trigonometric ratios

Write the exact value of each expression:

a $\sin 30°$ **b** $\cos 45°$ **c** $\tan 60°$ **d** $\sin 60°$

3 [Foundational] • Target: Trigonometric ratios

A right triangle has legs of length 7 and 24.



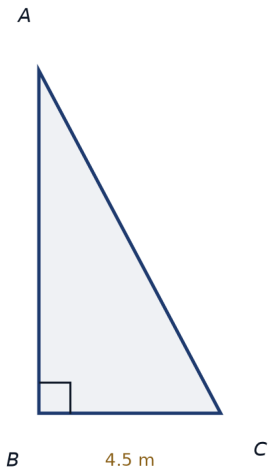
a Find the length of the hypotenuse.

b Find the measure of the angle opposite the side of length 7, to the nearest hundredth of a degree.

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4 [Foundational] • Target: Applications

A ladder leans against a wall, making an angle of $62°$ with the ground. The foot of the ladder is 4.5 m from the base of the wall. How high up the wall does the ladder reach? Round to two decimal places.



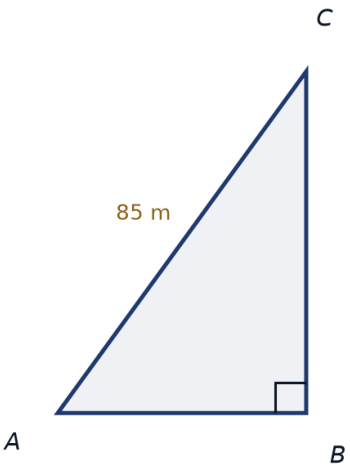
5 [Foundational] • Target: Applications

From the top of a cliff 50 m above sea level, the angle of depression to a boat is $18°$. How far is the boat from the base of the cliff? Round to one decimal place.



6 [Foundational] • Target: Applications

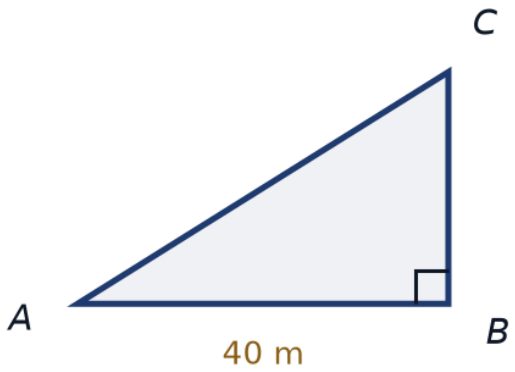
A kite string is 85 m long and makes an angle of $54°$ with the horizontal ground. Assuming the string is straight, find the height of the kite above the ground, to the nearest metre.



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7 [Foundational] • Target: Applications

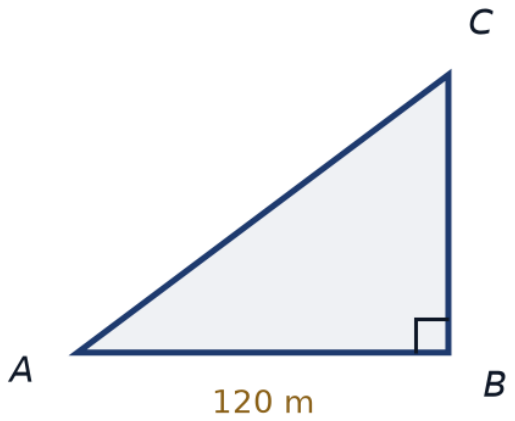
Two buildings are 40 m apart. From the top of the shorter building (height 25 m), the angle of elevation to the top of the taller building is $32°$. Find the height of the taller building, to one decimal place.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Applications

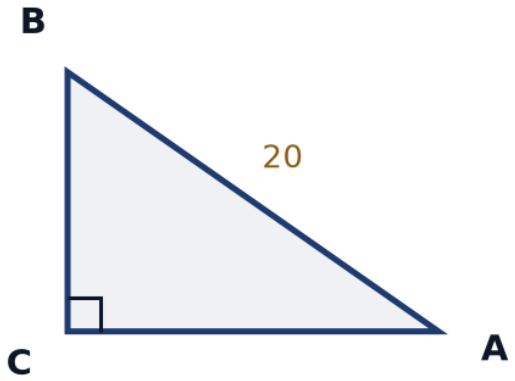
A surveyor stands 120 m from the base of a tower and measures the angle of elevation to the top as $37°$. Find the height of the tower, to the nearest tenth of a metre.



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9 [Intermediate] • Target: Solving right triangles

In right triangle ABC with the right angle at C, $A = 35°$ and the hypotenuse $AB = 20$. Find:



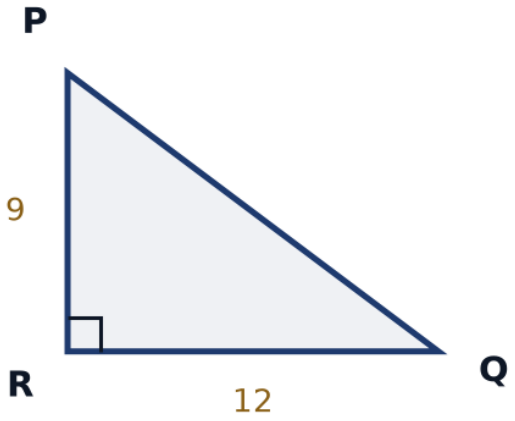
a side BC (opposite A)

b side AC (adjacent to A)

c angle B

10 [Intermediate] • Target: Solving right triangles

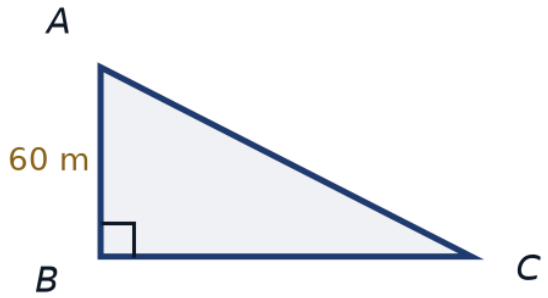
In right triangle PQR with the right angle at R, the two legs are $PR = 9$ and $QR = 12$. Find angle P (opposite QR) to the nearest tenth of a degree, and the length of the hypotenuse.



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11 [Intermediate] • Target: Angles of elevation and depression

A drone hovers at a height of 60 m. The angle of depression from the drone to a landmark on the ground is $27°$. Find the horizontal distance from the point directly below the drone to the landmark, to the nearest metre.

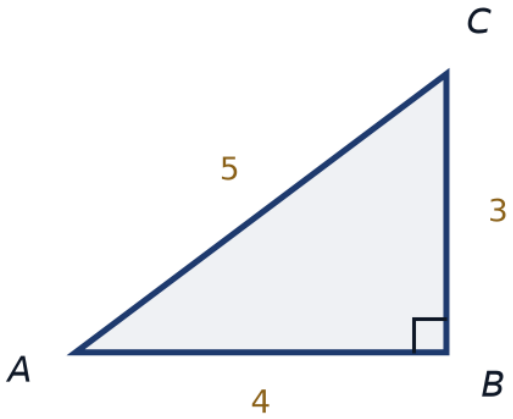


12 [Intermediate] • Target: Angles of elevation and depression

From a lighthouse 45 m tall, the angle of depression to a ship is $12°$, and a second ship is spotted further out with angle of depression $7°$. Find the distance between the two ships, to the nearest metre.

13 [Intermediate] • Target: The reciprocal trig ratios

For a right triangle with opposite $= 3$, adjacent $= 4$, hypotenuse $= 5$ (relative to angle θ), find:



a $\csc\theta$ **b** $\sec\theta$ **c** $\cot\theta$

14 [Intermediate] • Target: The reciprocal trig ratios

Explain why $\csc\theta = \frac{1}{\sin\theta}$, and use this to find $\csc(30°)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

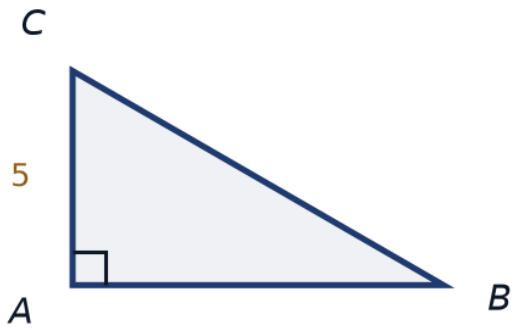
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15 [Challenging] • Target: Multi-step surveying problems • Cross-Lesson: 6.02 Polar Equations and Graphs

A surveyor at point A measures the angle of elevation to a mountain peak as $22°$. Walking 500 m directly toward the mountain to point B, the angle of elevation increases to $35°$. Find the height of the mountain, to the nearest metre. (Hint: let h be the height and x the distance from B to the base; set up two equations.)

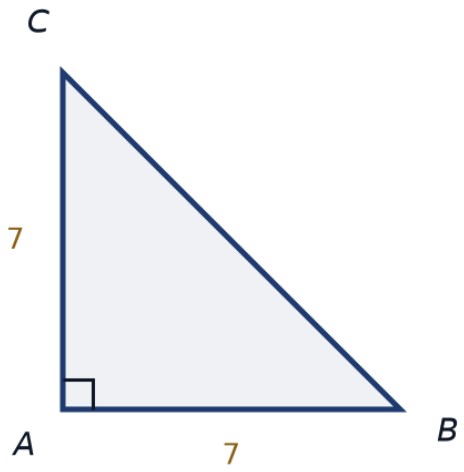
16 [Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

In a 30-60-90 triangle, the shortest side (opposite the $30°$ angle) has length 5. Find the lengths of the other two sides in exact form.



17 [Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 7.02 Logarithms and Their Properties

In a 45-45-90 triangle, each leg has length 7. Find the hypotenuse in exact form.



18 [Challenging] • Target: Trigonometric ratios beyond the first quadrant (preview) • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

A right triangle used to define $\sin\theta$ and $\cos\theta$ only works directly for acute angles θ. Explain why a different (coordinate-based) definition is needed to extend these ratios to obtuse or negative angles.

19 [Challenging] • Target: Solving right triangles • Cross-Lesson: 7.04 Applications: Growth and Decay

In ΔABC, $a = 8$, $b = 11$, and $A = 35^\circ$. Determine the number of possible triangles and find the measure of angle B for all valid cases to the nearest tenth of a degree.

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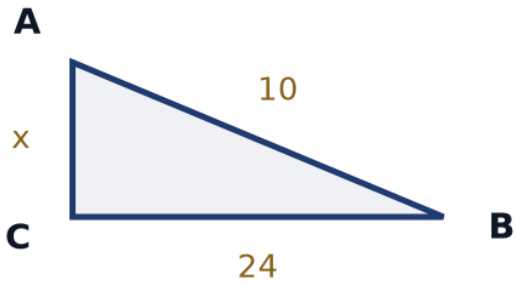
20 [Challenging] • Target: Angles of elevation and depression • Cross-Lesson: 8.01 Rational Functions and Asymptotes

A surveyor measures distances from an observation point C to two landmarks A and B: $AC = 140$ m and $BC = 210$ m, with included angle $\angle ACB = 78^\circ$. Find the distance AB between the landmarks to the nearest tenth of a meter, and compute the area of ΔABC to the nearest square meter.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Trigonometric ratios

In a right-angled triangle ABC, where the right angle is at C, if $AC = 10$ and $BC = 24$, find the length of the hypotenuse AB.



a. 25

b. 26

c. 27

d. 28

22 [Foundational] • Target: Applications

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

23 [Foundational] • Target: Solving right triangles

In triangle ABC, the lengths of sides a, b, and c are opposite to angles A, B, and C respectively. Given that $a = 7$, $b = 10$, and angle $C = 60^\circ$, find the exact length of side c.

a. $\sqrt{79}$

b. 9

c. $\sqrt{91}$

d. 10

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24 [Foundational] • Target: Angles of elevation and depression

If $\cos\theta = \frac{1}{2}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1}{4}$

b. $\frac{1}{2}$

c. $\frac{3}{5}$

d. $-\frac{1}{2}$

25 [Foundational] • Target: The reciprocal trig ratios

A complex number z is represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Given that $z^3 = 8i$, and z lies in the first quadrant such that its argument θ satisfies $0 < \theta < \frac{\pi}{2}$. Find the value of $r^2\cos\theta$.

a. 2

b. 4

c. $2\sqrt{2}$

d. 1

26 [Foundational] • Target: Multi-step surveying problems

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

27 [Foundational] • Target: The 30-60-90 and 45-45-90 special triangles

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$

d. $\frac{4}{5}$

28 [Foundational] • Target: Trigonometric ratios beyond the first quadrant (preview)

If $\sin\theta = \frac{5}{13}$ and θ is in Quadrant II, find $\cos\theta$.

a. $\frac{5}{13}$

b. $-\frac{12}{13}$

c. $-\frac{107}{130}$

d. $\frac{12}{13}$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Trigonometric ratios**

If $\cos\theta = -\frac{15}{17}$ and θ is in Quadrant III, find $\sin\theta$.

a. $-\frac{63}{170}$

b. $\frac{8}{17}$

c. $\frac{15}{17}$

d. $-\frac{8}{17}$

30 **[Intermediate]** • **Target: Applications**

If $\cos\theta = \frac{24}{25}$ and θ is in Quadrant IV, find $\sin\theta$.

a. $-\frac{7}{25}$

b. $\frac{24}{25}$

c. $-\frac{9}{50}$

d. $\frac{7}{25}$

31 **[Intermediate]** • **Target: Solving right triangles**

If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant I, find $\cos\theta$.

a. $\frac{4}{5}$

b. $\frac{3}{5}$

c. $\frac{9}{10}$

d. $-\frac{4}{5}$

32 **[Intermediate]** • **Target: Angles of elevation and depression**

If $\cos\theta = \frac{4}{5}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{4}{5}$

b. $-\frac{3}{5}$

c. $\frac{3}{5}$

d. $\frac{7}{10}$

33 **[Intermediate]** • **Target: The reciprocal trig ratios**

In a right-angled triangle, one of the acute angles measures 30°. If the side opposite this angle has a length of 5 units, what is the length of the hypotenuse?

a. 10 units

b. 5 units

c. $5\sqrt{3}$ units

d. 7.5 units

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34 [Intermediate] • Target: Multi-step surveying problems

In a right-angled triangle, one of the acute angles is 30°. What is the measure of the other acute angle?

a. 30°

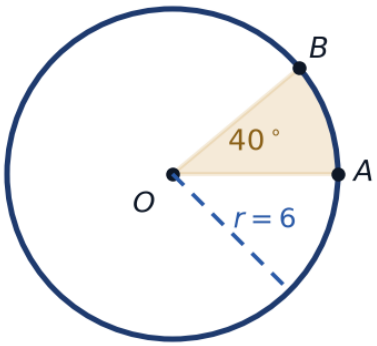
b. 45°

c. 60°

d. 90°

35 [Intermediate] • Target: The 30-60-90 and 45-45-90 special triangles

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?



a. $\cos\theta$

b. $\sin\theta$

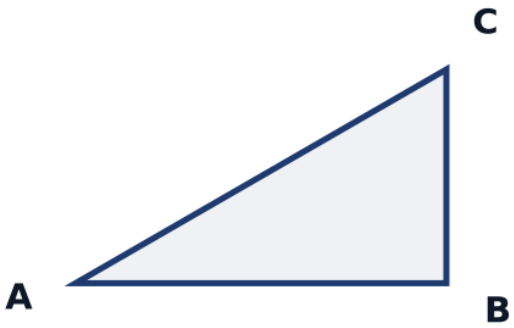
c. $\tan\theta$

d. 1

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36 [Intermediate] • Target: Trigonometric ratios beyond the first quadrant (preview)

In ΔABC, $\angle A = 30^\circ$, $\angle B = 90^\circ$, and side $a = 10$. Find side b (Law of Sines).



a. 10

b. 22

c. 5

d. 20

37 [Intermediate] • Target: Trigonometric ratios

Let the complex number z be represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Suppose z satisfies the equation $z^3 = -64i$. If z is restricted to the fourth quadrant (i.e., $\frac{3\pi}{2} < \arg(z) < 2\pi$), find the real part of z.

a. 4

b. -4

c. $2\sqrt{3}$

d. $-2\sqrt{3}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applications • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Consider a circle with the equation $x^2 + y^2 = 25$. A point $P(x_0, y_0)$ lies on this circle. A tangent line to the circle at point P intersects the x-axis at point A and the y-axis at point B. If the area of the triangle OAB, where O is the origin $(0, 0)$, is minimized, find the value of $x_0^2 + y_0^2$.

a. 25

b. 50

c. 100

d. 125

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39 [Challenging] • Target: Solving right triangles • Cross-Lesson: 7.04 Applications: Growth and Decay

Consider the complex number $z = \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{10}$. If z is expressed in polar form as $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $0 \leq \theta < 2\pi$, find the exact values of r and θ.

a. $r = 1, \theta = \frac{5\pi}{3}$

b. $r = 1, \theta = \frac{4\pi}{3}$

c. $r = 10, \theta = \frac{5\pi}{3}$

d. $r = \sqrt{3}, \theta = \frac{4\pi}{3}$

40 [Challenging] • Target: Angles of elevation and depression • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n?

a. $a_n = 5 + 3(n - 1)$

b. $a_n = 3n + 2$

c. $a_n = 5n + 3$

d. $a_n = 5 + 3n$

41 [Challenging] • Target: The reciprocal trig ratios • Cross-Lesson: 8.02 Graphing Rational Functions

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

a. 4

b. 5

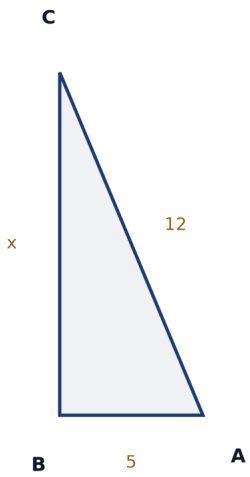
c. 6

d. 7

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42 [Challenging] • Target: Multi-step surveying problems • Cross-Lesson: 8.03 Introduction to Limits

In a right-angled triangle ABC, angle B is 90°. If $AB = 5$ and $BC = 12$, what is the length of the hypotenuse AC?



a. 10

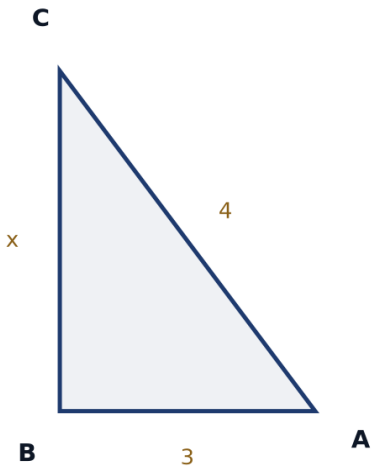
b. 11

c. 13

d. 17

43 [Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 9.01 Arithmetic Sequences and Series

In a right-angled triangle ABC, where angle B is 90 degrees, if AB = 3 and BC = 4, what is the length of the hypotenuse AC?



a. 7

b. 5

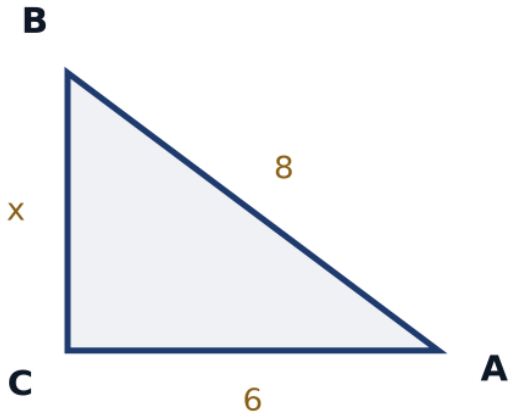
c. 12

d. 25

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44 [Challenging] • Target: Trigonometric ratios beyond the first quadrant (preview) • Cross-Lesson: 9.02 Geometric Sequences and Series

A right-angled triangle ABC is shown with the right angle at vertex C. The length of side AC is 8 units and the length of side BC is 6 units. What is the value of $\sin(\angle BAC)$?



a. $\frac{3}{5}$

b. $\frac{4}{5}$

c. $\frac{3}{4}$

d. $\frac{4}{3}$

45 [Challenging] • Target: Trigonometric ratios • Cross-Lesson: 9.03 The Binomial Theorem

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

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